

shyamsaran348@gmail.com

Portfolio

github.com/shyamsaran348

linkedin.com/in/shyam-saran

SUMMARY

Final-year IT student experienced in full-stack web development and AI-integrated systems. Skilled in backend development, RAG-based document search, and machine learning with Python.

EDUCATION

SSN College of Engineering, Chennai

Expected May 2027

B.Tech in Information Technology

CGPA: 7.9

Doveton Boys Higher Secondary School, Chennai

2023

Higher Secondary (Class XII)

90.2%

EXPERIENCE

Junior AI Analyst (Part-Time) A2000 Solutions

Jun 2026 – Present

- Fine-tuned a **Small Language Model (SLM)** using **QLoRA** to build an educational mentor AI assistant.
- Created synthetic **JSONL** training datasets and prompt templates to improve response quality.

Software Development Intern Highways Department, Govt. of Tamil Nadu

May – Jun 2026

- Built a full-stack portal using **React, FastAPI, and PostgreSQL** to streamline highway tender workflows.
- Implemented **JWT** authentication, hierarchical **RBAC**, and audit logs for multi-level approvals.

Web Developer Intern A2000 Solutions

Jun – Jul 2025

- Built a complete **LMS** using **Flask and PostgreSQL** with role-based access for students, instructors, and admins.
- Developed **REST APIs** for course management; optimized **SQL** queries to improve performance.

PROJECTS

VeriSource AI Python, FastAPI, PostgreSQL, ChromaDB, ONNX, React

GitHub

- Built a **RAG**-based platform for searching institutional rules/regulations documents.
- Implemented document retrieval using **ChromaDB** and **ONNX Runtime**, cutting memory usage by **60%**.

SDGSync – Enterprise Project Alignment Platform Python, FastAPI, PostgreSQL, React

GitHub

- Categorized project proposals against **UN SDGs** using a **BERT**-based classifier with RAG pipeline (**95%** accuracy).

AI-Powered Examination Portal (Ongoing) FastAPI, React.js, PostgreSQL

GitHub

- Examination portal with automated **MCQ generation**; building **PyTorch** proctoring engine for real-time behavioral monitoring.

SKILLS

Languages

Python, Java (Intermediate), C

Frameworks & APIs

FastAPI, Flask, React (Basic), RESTful APIs

AI Integration

Retrieval-Augmented Generation (RAG), QLoRA Fine-Tuning

Databases & Tools

PostgreSQL, MySQL, SQLite, Git, GitHub, VS Code

Concepts

DSA, OOP, DBMS, Operating Systems, Computer Networks

PUBLICATIONS

A Safety-Certified Framework for Maternal Triage: Integrating Conformal Prediction and Cost-Sensitive Ob-
jective Optimization

IEEE ICCVBIC 2026, Lalitpur, Nepal

Third Author

DOI

Reddit Rule-Violation Classification Using LLM-Assisted Data Augmentation

IEEE ICDSAAI 2026

Third Author

DOI

Explainable and Robust Heritage Intelligence Framework (ERHIF)

International Journal of Advanced Science and Engineering (IJASE), 2026

Fourth Author

DOI